

DRICS: Learning Disturbance Rejection in Command Space for Robust Quadruped Locomotion

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Fig. 1: DRICS under sustained interaction disturbances. A planar SRB model supports disturbance-aware command-space compensation through a frozen locomotion policy. Loaded-cart towing reduces forward-velocity tracking MAE by 68.0% indoors and 46.2% outdoors relative to Base.

Abstract—Quadruped locomotion policies achieve agile and stable motion under nominal conditions, but sustained interaction forces from payloads and resistive terrain can degrade velocity tracking or even stability. Retraining the entire locomotion policy for each interaction setting is costly and may degrade previously acquired locomotion skills. We present DRICS, a learned command-space disturbance-rejection method that augments a frozen command-conditioned locomotion policy. An upper-level compensator maps proprioception and an estimate of the planar disturbance wrench to bounded corrections of forward velocity, lateral velocity, and yaw rate. The corrected command is executed by the frozen base policy, whereas the reinforcement-learning objective remains defined with respect to the original user command. A single-rigid-body model defines a low-dimensional disturbance representation, and an extended-state observer provides force and yaw moment estimates. In simulation, DRICS with observer estimates reduces linear-velocity and yaw-rate mean absolute errors under disturbance by 69.1% and 62.9%, respectively, relative to Base, while preserving nominal performance with PPO and HIM base policies. Experiments on a Unitree Go2 validate the approach in indoor and outdoor loaded-cart towing, reducing forward-velocity mean absolute error by 68.0% and 46.2%, respectively, relative to towing with Base. These results demonstrate that command-space compensation is a modular alternative to re-learning joint-level locomotion control. An anonymous webpage with videos is available at <https://anonymous.4open.science/w/ICRA27-E4E6/>.

Index Terms—Quadruped Locomotion, Disturbance Rejection, Reinforcement Learning, Command Compensation, Extended-State Observer

I. INTRODUCTION

Quadruped robots are increasingly expected to operate while interacting with the environment, e.g., towing equipment, carrying payloads, or traversing vegetation and other resistive terrains. In such settings, the operator or planner specifies a desired planar velocity, but sustained external forces and moments cause tracking drift, gait degradation, and possible falls. Reliable autonomous operation therefore requires not only recovery from short impulses, but also rejection of persistent disturbances while preserving the requested motion state [1]–[3].

Model-based locomotion methods provide structured dynamics and explicit constraints through single-rigid-body (SRB), centroidal, or whole-body formulations, but their performance depends on model fidelity, contact information, and state or force estimation [4], [5]. Learning-based methods instead handle complex contact dynamics through large-scale offline training, with robustness commonly improved by domain randomization or online adaptation [6]–[8]. More recent disturbance-aware methods introduce external-force estimates or model-prediction errors into the control loop [9], [10]. However, they generally retrain or modify the joint-level policy, making it difficult to reuse a mature locomotion controller across new interaction settings.

A command-conditioned locomotion policy already maps a low-dimensional motion reference to coordinated whole-body actions [6], [8]. Hence, disturbance rejection need not be learned directly in the 12-dimensional joint-action space.

It can instead be learned at the 3D command interface: under disturbance, a virtual command bias can induce the frozen locomotion policy to generate the whole-body response needed to recover the original user-requested velocity. The compensated command is an internal control variable optimized only through its effect on tracking the original command. This places learned compensation at the command interface rather than in the model or joint action [11].

Based on this observation, we propose DRICS, a disturbance-aware command compensation layer for quadruped locomotion. We first derive a planar disturbance wrench from the SRB dynamics and append this wrench (its estimate in practice), to the locomotion observation. This disturbance augmentation can be used to train a unified joint-level robust policy. Our main method instead freezes the base policy and trains only an upper-level network that outputs bounded command corrections. During deployment, an extended-state observer (ESO) estimates the planar disturbance from measured base motion and reconstructed contact wrench [12]. The model provides a structured disturbance signal, while RL learns how this signal should reshape the command through the nonlinear base policy. The contributions of this work are threefold:

- (1) **Command-space disturbance rejection.** We introduce a hierarchical compensator that learns bounded three-dimensional command corrections through a frozen locomotion policy, while its reward remains defined against the original user command.
- (2) **Model-informed disturbance estimation.** We derive a planar body-frame disturbance representation from SRB dynamics and combine it with an ESO and asymmetric training to bridge privileged training signals and on-board disturbance estimates.
- (3) **Cross-policy and hardware validation.** We validate nominal preservation and disturbance rejection via fixed-condition comparisons, disturbance-velocity envelopes, and frozen PPO and HIM base policies, together with indoor/outdoor loaded-cart towing.

II. RELATED WORK

A. Model-Based Robust Legged Locomotion

SRB and centroidal models describe floating-base motion under contact wrenches. They support real-time quadratic programming and model-predictive control with friction and contact constraints [4]. Adaptive force-based control extends these models to payload and terrain uncertainty [5]. Such methods retain physical structure and may provide analytical guarantees, but require sufficiently accurate models, contact schedules, and state or force estimates. Active disturbance estimation and rejection offer another route by representing unknown dynamics as extended states [12]. In this work, the SRB model does not compute joint commands; it only defines and estimates low-dimensional disturbances.

B. Learning-Based Robust Locomotion

Deep RL enables quadrupeds to acquire contact-rich locomotion skills in massively parallel simulation and can

transfer them to hardware [6], [13]. Domain randomization exposes policies to diverse simulated parameters [14] and supports agile gaits over challenging terrains [15], [16]. However, its effectiveness depends on the training distribution and policy capacity [17]: narrow randomization may miss relevant conditions, whereas broad randomization makes optimization harder, and results in conservative performance.

Adaptation methods address changes in terrain, payload, and external forces. RMA infers environmental properties from state-action histories [7], HIM uses proprioceptive robot responses for agile blind locomotion under changing conditions [8], and PA-LOCO combines privileged feature decoupling with residual learning under force perturbations [18]. Because these methods infer disturbances from their effects on robot motion, some degradation may occur before sufficient information is collected. Our method instead supplies an explicit planar disturbance estimate to a pretrained policy while freezing its joint-level controller.

C. Hybrid and Disturbance-Aware Learning

Hybrid model-learning methods retain nominal controllers while learning compensation for uncertainties. Mismatch compensation combines RL with MPC through compensatory actions or prediction errors [11]. For legged robots, RL2AC adds online adaptive torque compensation [19], whereas RAMBO combines whole-body feedforward torques with learned corrections [2]. They preserve control structures but apply learning mainly in action or torque space.

Disturbance-aware methods expose estimated forces or model-prediction errors to the controller. Seeing-Eye estimates force direction for tug-responsive navigation [1], and HAC-LOCO uses estimated forces for command-level compliance [3]. DATT replaces privileged disturbance input with an adaptive estimate at deployment [9], while NeuralIMC feeds model-prediction error to the policy [10]. DRICS in this work combines both directions: an SRB model and ESO estimate planar disturbances, while a learned module modifies only a frozen policy's three-dimensional command, leaving its joint-level controller unchanged.

III. PRELIMINARIES

A. Disturbed Single-Rigid-Body Dynamics

To characterize external-interaction effects on quadruped motion, we adopt an SRB model. It represents the robot as a rigid body centered at its center of mass (CoM), neglecting limb inertia while retaining the ground-reaction forces and external-interaction wrenches acting on the body [2], [4], [5]. Let $\{W\}$ and $\{B\}$ denote the inertial world and robot body frames. The rotation $\mathbf{R}_{WB} \in SO(3)$ maps vectors from $\{B\}$ to $\{W\}$. Expressing linear and angular dynamics in $\{W\}$ yields the disturbed Newton-Euler equations

$$m\dot{\mathbf{v}}_c^W = m\mathbf{g}^W + \sum_{i=1}^{n_c} \mathbf{f}_i^W + \mathbf{f}_d^W, \quad (1a)$$

$$\mathbf{I}^W \dot{\boldsymbol{\omega}}^W + \boldsymbol{\omega}^W \times (\mathbf{I}^W \boldsymbol{\omega}^W) = \sum_{i=1}^{n_c} \mathbf{r}_i^W \times \mathbf{f}_i^W + \boldsymbol{\tau}_d^W. \quad (1b)$$

where m is the total mass; $\mathbf{v}_c^W$ and $\boldsymbol{\omega}^W$ are the CoM linear and angular velocities; $\mathbf{g}^W$ is gravity; and $\mathbf{I}^W = \mathbf{R}_{WB} \mathbf{I}^B \mathbf{R}_{WB}^\top$ is the world-frame inertia. The force $\mathbf{f}_i^W$ acts at the i th contact, n_c is the number of active contacts, and $\mathbf{r}_i^W = \mathbf{p}_i^W - \mathbf{p}_c^W$ is its CoM-relative moment arm. All contact quantities in (1) are expressed in the same frame.

An external interaction force $\mathbf{F}_{\text{ext}}^W$ applied at $\mathbf{p}_{\text{ext}}^W$ produces the equivalent CoM disturbance wrench

$$\mathbf{f}_d^W = \mathbf{F}_{\text{ext}}^W, \quad (2a)$$

$$\boldsymbol{\tau}_d^W = \mathbf{r}_{\text{ext}}^W \times \mathbf{F}_{\text{ext}}^W, \quad \mathbf{r}_{\text{ext}}^W = \mathbf{p}_{\text{ext}}^W - \mathbf{p}_c^W. \quad (2b)$$

We project (1) onto the planar command space and express the reduced dynamics in the body frame, aligning the model with the policy's body-frame commands and proprioceptive observations. Define $\boldsymbol{\nu} = [v_x^B, v_y^B, \omega_z^B]^\top$, $\mathbf{d}^B = [f_x^B, f_y^B, \tau_z^B]^\top$. Under the near-level planar approximation, we retain horizontal translation, yaw, and the rotating-frame transport term while neglecting higher-order gyroscopic terms, giving

$$\mathbf{M}_p \dot{\boldsymbol{\nu}} = \mathbf{u}_c^B + \mathbf{d}^B - \begin{bmatrix} m\omega_z^B \mathbf{J} \mathbf{v}_{xy}^B \\ 0 \end{bmatrix}, \quad \mathbf{M}_p = \text{diag}(m, m, I_z), \quad (3)$$

where $\mathbf{J} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, $\mathbf{v}_{xy}^B = [v_x^B, v_y^B]^\top$, and $\mathbf{u}_c^B$ is the planar resultant ground-contact wrench. The disturbance $\mathbf{d}^B = [f_x^B, f_y^B, \tau_z^B]^\top$ contains the horizontal interaction force and yaw moment. Eq. (3) thus defines a compact disturbance representation aligned with the policy command, without reproducing the full hybrid limb dynamics.

With ψ denoting base yaw, the world- and body-frame representations of the same disturbance satisfy

$$\mathbf{d}^B = \mathbf{T}(\psi) \mathbf{d}^W, \quad \mathbf{T}(\psi) = \text{blkdiag}(\mathbf{R}_z^\top(\psi), 1), \quad (4)$$

where blkdiag forms a block-diagonal matrix from its arguments. We use $\mathbf{d}^B$ as the policy observation so that disturbance directions remain consistently aligned with the forward, lateral, and yaw commands.

B. Disturbance-Augmented Policy Formulation

To enable the locomotion controller to actively reject external interactions rather than relying solely on implicit policy robustness, we introduce the external disturbance as an explicit signal in the observation space. By exposing the disturbance direction and magnitude to the policy, the augmented observation allows a unified disturbance-aware policy to adapt its joint-level actions to counteract the disturbance and maintain the desired motion state. A standard PPO locomotion policy uses a 48-dimensional observation

$$\mathbf{o}_t^b = [\mathbf{v}_t^B, \boldsymbol{\omega}_t^B, \mathbf{g}_t^B, \mathbf{c}_t^0, \mathbf{q}_t - \mathbf{q}_0, \dot{\mathbf{q}}_t, \mathbf{a}_{t-1}], \quad (5)$$

where $\mathbf{c}_t^0 = [v_x^d, v_y^d, \omega_z^d]^\top$ is the original user command. We form a robust augmented observation

$$\mathbf{o}_t^+ = [\mathbf{o}_t^b, \tilde{\mathbf{d}}_t^B], \quad \tilde{\mathbf{d}}_t^B = \mathbf{S}_d \hat{\mathbf{d}}_t^B, \quad (6)$$

where $\mathbf{S}_d$ normalizes the force and moment channels. In simulation, $\hat{\mathbf{d}}^B$ can be the applied ground-truth wrench; at deployment it is supplied by the ESO in Section IV-D.

The direct use of (6) is a unified disturbance-aware policy

$$\mathbf{a}_t \sim \pi_u(\cdot \mid \mathbf{o}_t^+), \quad \pi_u : \mathbb{R}^{51} \rightarrow \mathbb{R}^{12}. \quad (7)$$

The resulting unified policy serves as a strong end-to-end disturbance-aware baseline. Although it directly exploits disturbance information in the joint-action space, it requires retraining the locomotion policy. In contrast, our DRICS retains the pretrained base policy and learns only at its command interface.

IV. DRICS FRAMEWORK

DRICS is designed to exploit the disturbance-rejection capability already latent in a command-conditioned locomotion policy, without retraining that policy in the joint-action space. Instead, we learn a command-level disturbance-rejection policy that converts proprioception and explicit disturbance information into a compensating velocity command. The compensation reshapes the reference seen by the existing controller, enabling the robot to actively counteract external interactions while preserving the locomotion skill encoded by the pretrained policy.

A. Hierarchical Policy Composition

Let π_b denote a pretrained command-conditioned base policy. Its parameters remain frozen throughout DRICS training. On top of π_b , we introduce a feed-forward multilayer perceptron π_c that maps the 51-dimensional disturbance-augmented observation to a 3-D command correction:

$$\Delta \mathbf{c}_t = \pi_c(\mathbf{o}_t^+), \quad \pi_c : \mathbb{R}^{51} \rightarrow \mathbb{R}^3. \quad (8)$$

The complete hierarchical policy is

$$\mathbf{c}_t^+ = \text{proj}_{\mathcal{C}_b}(\mathbf{c}_t^0 + \Delta \mathbf{c}_t), \quad (9a)$$

$$\mathbf{a}_t \sim \pi_b(\cdot \mid \mathcal{T}(\mathbf{o}_t^b, \mathbf{c}_t^+)), \quad (9b)$$

where $\mathcal{T}$ replaces only the command slice of the base observation and $\mathcal{C}_b$ denotes the command support of the pretrained policy. Because DRICS relies on the motion capability already represented by π_b , the original-command sampling set $\mathcal{C}_0$ and admissible correction set $\Delta \mathcal{C}$ are selected such that their composition remains within that capability, i.e., $\mathcal{C}_0 \oplus \Delta \mathcal{C} \subseteq \mathcal{C}_b$. The remaining margin and final projection keep the command seen by π_b inside its training support.

B. Cross-Layer Policy Optimization

The important distinction in DRICS is that the base policy tracks $\mathbf{c}_t^+$, but the task reward is evaluated against $\mathbf{c}_t^0$:

$$r_t^v = \exp\left(-\frac{\|\mathbf{v}_{xy,t}^B - \mathbf{c}_{xy,t}^0\|_2^2}{\sigma_v}\right), \quad (10a)$$

$$r_t^\omega = \exp\left(-\frac{(\omega_{z,t} - c_{\omega,t}^0)^2}{\sigma_\omega}\right), \quad (10b)$$

$$r_t = w_v r_t^v + w_\omega r_t^\omega + r_t^{\text{loco}}. \quad (10c)$$

where r_t^{loco} contains the base locomotion regularizers, including vertical motion, roll/pitch rate, torque, action-rate, collision, and joint-limit terms. Consequently, a command

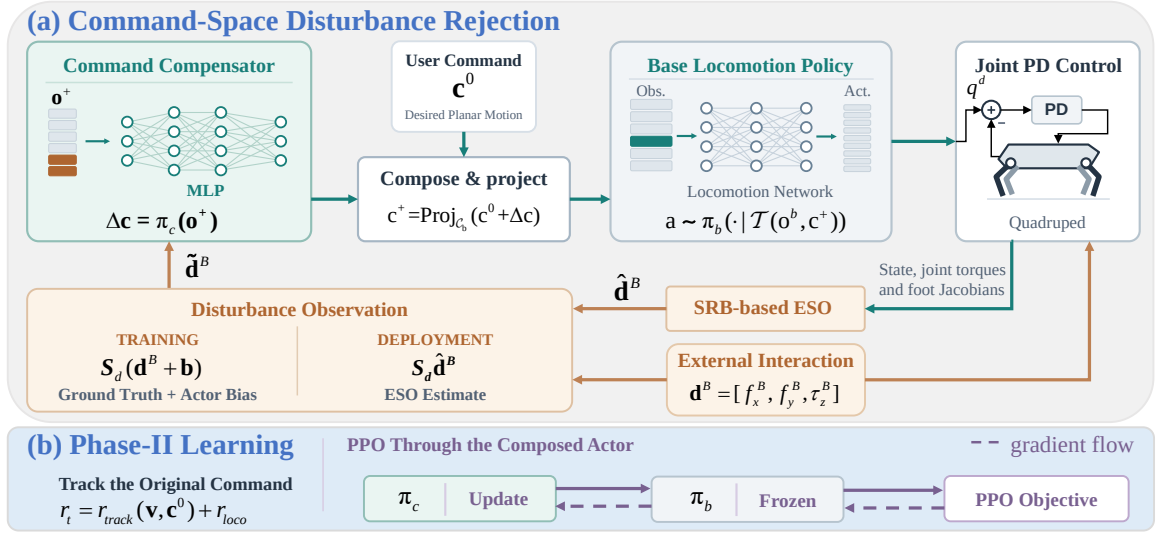


Fig. 2: **Overview of the proposed DRICS.** (a) A command compensator reshapes the velocity command to the base policy. Disturbance observations use biased ground truth during training and ESO estimates at deployment. (b) PPO optimizes tracking of the original command: actor gradients traverse the frozen base and update only the compensator; the asymmetric critic receives ground-truth disturbance.

correction is useful only if its closed-loop execution through π_b reduces error relative to the original task.

We optimize π_c with PPO [20]. Let θ denote its trainable parameters. The resulting composed stochastic actor is

$$\pi_\theta(\cdot | \mathbf{o}_t^+) = \mathcal{N}(\mu_b(\mathcal{T}(\mathbf{o}_t^b, \mathbf{c}_t^+(\theta))), \text{diag}(\sigma_b^2)). \quad (11)$$

During backpropagation, gradients traverse the differentiable input-output map of π_b to assign credit to the upstream command correction, but the parameters of π_b are not updated. Among the actor networks, only π_c is optimized, while the value function follows the standard PPO update. This cross-layer optimization lets the compensator learn from the final joint-level locomotion outcome, preserving the pretrained controller and avoiding relearning the gait skill.

C. Disturbance Curriculum and Asymmetric Training

Training proceeds in two stages. Stage one trains or imports the nominal base policy π_b without sustained interaction disturbances. Stage two freezes π_b and trains π_c while continuously applying the planar three-DoF wrench $\mathbf{d}^B = [f_x^B, f_y^B, \tau_z^B]^\top$ to the robot body.

We use a disturbance curriculum that first exposes the policy to constant body-frame wrenches and then progressively increases the proportion and variation rate of time-varying interactions. The time-varying set includes world-locked constant forces represented in the body frame, independently rotating planar forces, and multi-sinusoidal force and yaw-moment profiles. Constant-disturbance samples remain in the curriculum, so the policy first acquires steady-state rejection and subsequently extends it to interactions whose direction and magnitude evolve over time.

The actor is trained with the ground-truth disturbance, whereas deployment replaces this privileged signal with an online estimate via the ESO in Section IV-D. To improve robustness to the resulting estimation bias, we use asymmetric

disturbance observations during training:

$$\mathbf{o}_t^{+,a} = [\mathbf{o}_t^b, \mathbf{S}_d(\mathbf{d}_t^B + \mathbf{b})], \quad (12a)$$

$$\mathbf{o}_t^{+,v} = [\mathbf{o}_t^b, \mathbf{S}_d \mathbf{d}_t^B]. \quad (12b)$$

where $\mathbf{b}$ is an episode-wise random bias, while superscripts a and v denote actor and value-function observations. Thus, the actor learns command compensation under degraded disturbance information, while the critic retains clean ground-truth information to stabilize learning. At deployment, the same actor accepts the estimated disturbance without changing the hierarchical control interface.

D. ESO-Based Disturbance Estimation

To provide disturbance estimation, we formulate the ESO in the body frame. Let the planar motion state be

$$\mathbf{x}^B = [v_x^B, v_y^B, \omega_z^B]^\top, \text{ then } \dot{\mathbf{x}}^B = \mathbf{u}_a^B + \boldsymbol{\delta}^B, \quad (13)$$

where $\boldsymbol{\delta}^B = \mathbf{M}_p^{-1} \mathbf{d}^B$ is the generalized acceleration disturbance. From (3), the known body-frame input is

$$\mathbf{u}_a^B = \mathbf{M}_p^{-1} \left(\mathbf{u}_c^B - \begin{bmatrix} m\omega_z^B \mathbf{J} \mathbf{v}_{xy}^B \\ 0 \end{bmatrix} \right). \quad (14)$$

Thus, the ground-contact wrench and the rotating-frame transport term are treated as known model inputs, while the external interaction is represented by the extended state. A second-order linear ESO is

$$\mathbf{e} = \mathbf{z}_1 - \mathbf{x}^B, \quad (15a)$$

$$\dot{\mathbf{z}}_1 = \mathbf{u}_a^B + \mathbf{z}_2 - \mathbf{B}_1 \mathbf{e}, \quad (15b)$$

$$\dot{\mathbf{z}}_2 = -\mathbf{B}_2 \mathbf{e}, \quad (15c)$$

$$\hat{\mathbf{d}}^B = \mathbf{M}_p \mathbf{z}_2, \quad (15d)$$

with diagonal bandwidth parameterization

$$\mathbf{B}_1 = 2 \text{diag}(\omega_o), \quad \mathbf{B}_2 = \text{diag}(\omega_o)^2, \quad (16)$$

where $\mathbf{z}_1$ estimates the planar motion state, $\mathbf{z}_2$ estimates the generalized disturbance acceleration, and $\omega_o =$

$[\omega_{o,x}, \omega_{o,y}, \omega_{o,z}]^\top$ contains the observer bandwidths for the two translational channels and yaw. Larger bandwidths improve tracking but amplify noise, contact impacts, and unmodeled dynamics; smaller values smooth estimates but increase phase lag. The nominal contact wrench can be obtained from measured or commanded joint torques. For a stance leg with foot Jacobian $\mathbf{J}_i$, we use

$$\hat{\mathbf{f}}_i^B \approx \mathbf{J}_i^{-\top} \boldsymbol{\tau}_i, \quad \hat{\mathbf{u}}_c^B = \sum_i \begin{bmatrix} \hat{\mathbf{f}}_{i,xy}^B \\ (\mathbf{r}_i^B \times \hat{\mathbf{f}}_i^B)_z \end{bmatrix}. \quad (17)$$

This approximation omits limb inertia, gravity, friction, and actuator dynamics; their residual effects are absorbed into the estimated total disturbance. Contact masking is obtained from foot-contact estimates.

V. QUANTITATIVE SIMULATION COMPARISON

A. Experimental Setup

Simulation and training. We simulate a 15 kg Unitree Go2 in Isaac Gym. Physics and joint PD control run at 200 Hz, with policy inference at 50 Hz. Joint-position targets use $K_p = 30$ N m/rad and $K_d = 0.75$ N m s/rad. All PPO actors and critics use hidden dimensions (512, 256, 128) with ELU activations. Base and DR-Base map 48-dimensional observations to 12 joint-position offsets, Unified maps 51-dimensional observations to the same action space, and π_c maps 51-dimensional observations to three command corrections. Training uses 4096 parallel environments, 48-step rollouts, and 2000 PPO updates, with an adaptive learning rate initialized at 10^{-3} .

Command and disturbance ranges. The base-policy support is $v_x, v_y \in [-1.5, 1.5]$ m/s and $\omega_z \in [-2, 2]$ rad/s. During compensation training, each original-command channel is sampled from $[-1, 1]$, in m/s for translation and rad/s for yaw. Reward widths are $\sigma_v = \sigma_\omega = 0.25$ in Stage I and $\sigma_v = \sigma_\omega = 0.1$ in Stage II; DR-Base and Unified use the Stage-II command range. Corrections are limited to (0.5, 0.5, 1.0) in m/s, m/s, and rad/s. The sustained body-frame wrench is sampled as

$$f_x^B, f_y^B \sim \mathcal{U}(-30, 30) \text{ N}, \quad \tau_z^B \sim \mathcal{U}(-5, 5) \text{ N m}. \quad (18)$$

Disturbance curriculum. Time-varying disturbances are introduced after 500 updates, and the final mixture is reached at 1500 updates. At the final level, 75% of environments use time-varying profiles: one third are world-locked, with the remainder divided between independently rotating and multi-sinusoidal profiles. Rotating and sinusoidal frequencies are sampled from $[0.1, 1.0]$ Hz, while sinusoidal amplitudes increase to 35% of each channel range.

Randomization and ESO. Training randomizes friction over $[0.2, 1.5]$, added base mass over $[-1.5, 1.5]$ kg, and motor-strength scale over $[0.9, 1.1]$, together with joint zero offsets, initial states, and impulsive pushes. The body-frame ESO uses $\omega_o = [15, 15, 5]$ rad/s, $m = 15$ kg, and $I_z = 0.5$ kg m². Its contact input is reconstructed from commanded joint torques and foot Jacobians using (17); the estimated wrench is clipped to the training support before normalization.

Evaluation protocol. Unless noted otherwise, evaluation disables domain randomization. Episodes last 20 s, with constant commands and disturbances except in time-varying tests. Linear and angular tracking errors are

$$E_v = \frac{1}{T} \sum_{t=1}^T \|\mathbf{v}_{xy,t}^B - \mathbf{c}_{xy,t}^0\|_2, \quad E_\omega = \frac{1}{T} \sum_{t=1}^T |\omega_{z,t} - c_{\omega,t}^0|.$$

We report tracking MAE and survival, defined as the fraction of episodes completed without a fall or early termination.

B. Simulation Results

We compare Base, DR-Base, Unified, and DRICS, whose structural differences are summarized in Tab. I. Unified and DRICS use either ground-truth (GT) disturbance observations or ESO estimates at evaluation.

TABLE I: Structural comparison of evaluated policies.

Method	Explicit $\mathbf{d}$	Retrain base	Output space
Base	No	No	12-D joint
DR-Base	No	Yes	12-D joint
Unified	Yes	Yes	12-D joint
DRICS (ours)	Yes	No	3-D command

1) **Random-scenario performance:** We first evaluate robustness across random combinations of motion commands and external disturbances while checking whether nominal locomotion is preserved. We sample $v_x^d, v_y^d, \omega_z^d \in [-1, 1]$ and disturbances from (18). Each trial uses 50 environments with 20 episodes per environment, and Tab. II reports results across three random seeds.

Under disturbance, Base linear and yaw MAEs increase by $3.9\times$ and $5.2\times$, respectively, while survival decreases by 6.1 percentage points. DRICS-GT reduces these errors by 70.4% and 69.0%. Using ESO estimates retains most of this benefit: DRICS-ESO achieves reductions of 69.1% and 62.9% and increases survival from 93.8% to 99.3%. It also preserves Base-level nominal tracking and survival. Unified is modestly more accurate under disturbance because it can reshape the complete joint-level policy, whereas DRICS retains the pretrained locomotion skill at the cost of a small residual tracking gap.

2) **Fixed-Condition Comparison:** We then evaluate arc motion (1, 0, 1) and diagonal motion (1, 1, 0) under the constant body-frame disturbance $\mathbf{d}^B = (20, 20, 4)$. Tab. III reports mean $\pm$ standard deviation over 50 parallel environments, with the final column averaging the relative reductions in E_v and E_ω across both conditions.

DR-Base shows that disturbance randomization improves implicit robustness but remains less effective than explicit disturbance feedback. Across the four tracking metrics, DRICS-ESO reduces errors by 70.0%–81.6%, yielding a mean reduction of 75.8%, close to 79.3% for Unified-ESO while retaining the frozen base policy. Its mean absolute command corrections are (0.155, 0.169, 0.265) and (0.112, 0.116, 0.363) for arc and diagonal motion, respectively, in units of m/s, m/s, and rad/s, remaining within the available command margins.

TABLE II: Random-scenario performance (mean $\pm$ SD across seeds).

Method	No Disturbance			With Disturbance		
	E_v [m/s] $\downarrow$	E_ω [rad/s] $\downarrow$	Surv. [%] $\uparrow$	E_v [m/s] $\downarrow$	E_ω [rad/s] $\downarrow$	Surv. [%] $\uparrow$
Base	0.063 $\pm$ 0.006	0.038 $\pm$ 0.005	99.9 $\pm$ 0.1	0.243 $\pm$ 0.003	0.197 $\pm$ 0.008	93.8 $\pm$ 0.8
DRICS-GT	0.052 $\pm$ 0.005	0.039 $\pm$ 0.005	99.9 $\pm$ 0.1	0.072 $\pm$ 0.005	0.061 $\pm$ 0.005	99.5 $\pm$ 0.2
DRICS-ESO	0.052 $\pm$ 0.005	0.039 $\pm$ 0.005	99.9 $\pm$ 0.1	0.075 $\pm$ 0.005	0.073 $\pm$ 0.005	99.3 $\pm$ 0.2
Unified-GT	0.056 $\pm$ 0.003	0.044 $\pm$ 0.004	100.0	0.066 $\pm$ 0.004	0.053 $\pm$ 0.004	99.9 $\pm$ 0.1
Unified-ESO	0.057 $\pm$ 0.003	0.045 $\pm$ 0.003	100.0	0.066 $\pm$ 0.004	0.064 $\pm$ 0.003	99.9 $\pm$ 0.1

TABLE III: Fixed-condition comparison under $\mathbf{d}^B = (20, 20, 4)$. Reduction is averaged over the four errors relative to the disturbed Base.

Method	Arc (1, 0, 1)		Diagonal (1, 1, 0)		Reduction [%]
	E_v $\downarrow$	E_ω $\downarrow$	E_v $\downarrow$	E_ω $\downarrow$	
Base, no disturbance	0.059 $\pm$ 0.003	0.038 $\pm$ 0.002	0.068 $\pm$ 0.003	0.028 $\pm$ 0.002	–
Base	0.295 $\pm$ 0.003	0.260 $\pm$ 0.002	0.303 $\pm$ 0.004	0.353 $\pm$ 0.003	0.0
DR-Base	0.080 $\pm$ 0.002	0.112 $\pm$ 0.001	0.122 $\pm$ 0.004	0.097 $\pm$ 0.001	65.5
DRICS-GT	0.073 $\pm$ 0.002	0.055 $\pm$ 0.002	0.077 $\pm$ 0.003	0.066 $\pm$ 0.003	77.5
DRICS-ESO	0.079 $\pm$ 0.002	0.056 $\pm$ 0.002	0.091 $\pm$ 0.003	0.065 $\pm$ 0.003	75.8
Unified-GT	0.075 $\pm$ 0.002	0.047 $\pm$ 0.003	0.062 $\pm$ 0.003	0.048 $\pm$ 0.003	80.6
Unified-ESO	0.078 $\pm$ 0.002	0.058 $\pm$ 0.003	0.063 $\pm$ 0.003	0.047 $\pm$ 0.005	79.3

TABLE IV: Mean gain across the disturbance–velocity grids. Large-disturbance columns use $|f_x^B|$ or $|f_y^B| \geq 20$ N, or $|\tau_z^B| = 5$ N m.

Grid	GT all	ESO all	GT large	ESO large
$v_x - f_x$	55.8%	53.5%	68.8%	65.0%
$v_y - f_y$	59.0%	57.9%	72.4%	70.5%
$\omega_z - \tau_z$	60.0%	57.7%	82.6%	81.8%

3) **Disturbance–velocity envelopes:** We evaluate the operating envelope by varying each command–disturbance pair. For the translational grids, the commanded velocity is sampled at $\{-1, -0.5, 0, 0.5, 1\}$ m/s and the corresponding body-frame force at $\{-30, -20, -10, 0, 10, 20, 30\}$ N, yielding 35 points per grid. For yaw, $\omega_z^d \in \{-1, -0.5, 0, 0.5, 1\}$ rad/s and $\tau_z^B \in \{-5, -2, 0, 2, 5\}$ N m yield 25 points, with $v_x^d = 0.5$ m/s. Each point uses 50 environments and five episodes. Fig. 3 presents the ESO results, while Tab. IV compares GT and ESO gains, defined as $(E_{\text{Base}} - E_{\text{DRICS}})/E_{\text{Base}}$.

Across the translational grids, DRICS-ESO achieves mean gains of 53.5%–57.9%, increasing to 65.0%–70.5% at large forces. It improves yaw tracking at 20 of 25 points and reaches an average gain of 81.8% at $|\tau_z^B| = 5$ N m. Small or zero yaw disturbances can yield negligible or slightly negative gains because Base error is already low and ESO transients may trigger unnecessary corrections. Overall, ESO retains most of the gain achieved with ground-truth disturbance observations.

4) **Generalization across base policies:** To test modularity across locomotion policies, we apply DRICS to trained PPO and HIM base policies [8]. Each base policy remains frozen while only its corresponding compensator is trained. For HIM, the compensator uses the velocity estimate produced by its proprioceptive-history pipeline rather than ground-truth base velocity. Tab. V reports results under the same random-scenario protocol used above.

Under disturbance, HIM-DRICS-ESO reduces linear and yaw MAE by 68.5% and 60.5%, respectively, and increases survival from 92.9% to 99.3%, while preserving nominal performance. Its disturbed linear error remains higher than that of the PPO-based DRICS-ESO, consistent with the

TABLE V: Generalization to a frozen HIM base policy.

Method	No Disturbance			With Disturbance		
	E_v $\downarrow$	E_ω $\downarrow$	Surv. $\uparrow$	E_v $\downarrow$	E_ω $\downarrow$	Surv. $\uparrow$
HIM-Base	0.068	0.038	100.0	0.304	0.189	92.9
HIM-DRICS-GT	0.052	0.041	100.0	0.086	0.066	98.8
HIM-DRICS-ESO	0.052	0.041	100.0	0.096	0.075	99.3

absence of direct base-velocity observations, whereas yaw error and survival are comparable. These results demonstrate that DRICS can reuse independently developed command-conditioned locomotion policies without modifying their joint-level controllers.

VI. REAL-WORLD EXPERIMENTS

A. Hardware and Evaluation Protocol

We deploy the controllers on a Unitree Go2 equipped with a Jetson AGX Orin computer and a Livox MID-360 LiDAR. FAST-LIO2 estimates linear velocity and position at 50 Hz, while angular velocity is obtained from the onboard IMU. Policy inference and joint PD control run at 50 and 500 Hz, respectively. The robot tows a loaded camping cart through a rope, producing sustained disturbances from rolling resistance, wheel steering, and changing rope direction. DRICS uses ESO estimates without modifying the pretrained base policy.

We compare unloaded Base, towing Base, and towing DRICS in indoor and outdoor straight-motion tasks with command $(v_x^d, 0, 0)$. Each towing and outdoor unloaded condition contains two trials; one unloaded trial is available indoors. Per-axis MAE is computed from unsmoothed measurements over 2–10 s after command onset and reported as trial mean and sample standard deviation. Moving averages are used only for visualization.

B. Indoor Asymmetric-Load Towing

The experiment is conducted in a 2.2 m-wide marble-floor corridor (Fig. 5). The cart carries a second Go2 and 9 L of water, including 3 L placed on its right side. The robot is commanded to track a 0.5 m/s forward command for 20 s. Fig. 4 and Tab. VI show that towing decreases the Base mean

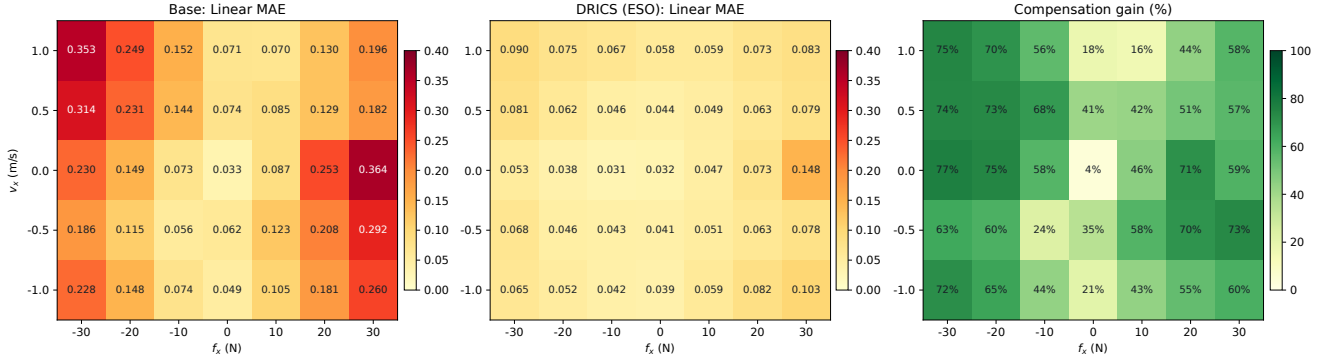


Fig. 3: **Disturbance-velocity envelope for the forward channel** using ESO estimation. The v_x - f_x grid compares the Base and DRICS tracking MAEs with the relative compensation gain. Results for the lateral/yaw channels are on the anonymous project webpage.

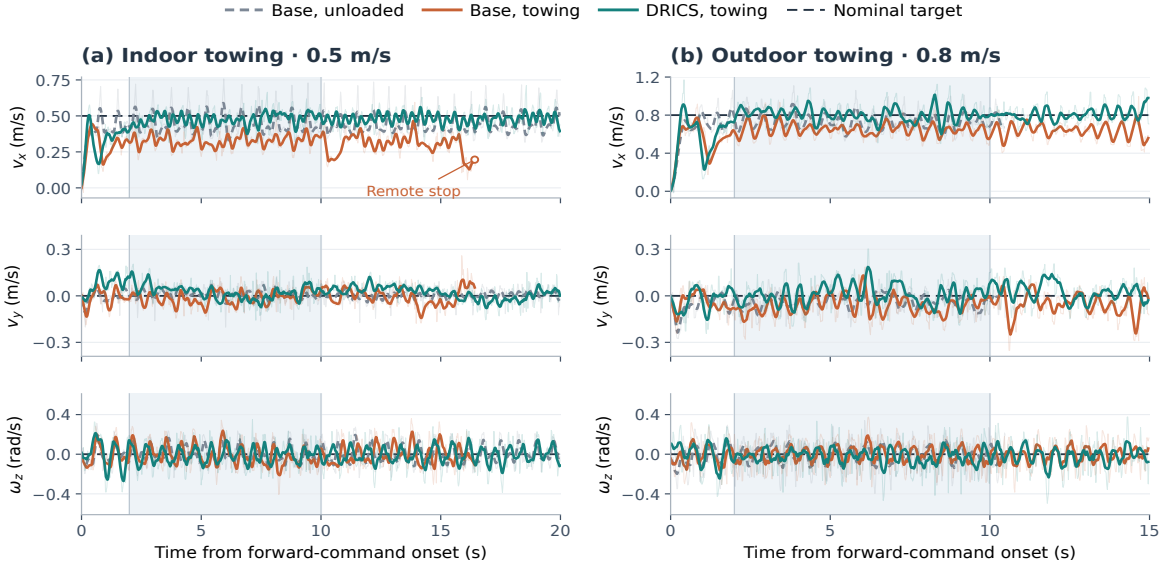


Fig. 4: **Body-frame velocities in real-world towing** with a frozen locomotion policy. (a) indoor and (b) outdoor trials. Faint and dark traces show raw measurements and 0.2 s moving averages, respectively. Shading marks the 2–10 s analysis interval used in Tab. VI. Curves end at the final recorded sample without extrapolation; the outdoor unloaded trace ends at 10.5 s. The open marker denotes a recorded remote stop rather than a precisely synchronized wall-contact time.

forward velocity from 0.442 to 0.335 m/s. DRICS restores it to 0.479 m/s and reduces forward MAE from 0.166 to 0.053 m/s, a 68.0% reduction. Despite similar lateral-velocity MAEs, both Base trials drift toward the right wall. At 10 s, their initial-heading-aligned lateral displacements are -0.539 and -0.430 m, versus 0.033 and 0.102 m for DRICS. Fig. 6 shows that the compensator supplies a mean forward reference of 0.646 m/s over 2–10 s, enabling recovery toward the original 0.5 m/s target.

C. Outdoor Towing on a Rubber Surface

The outdoor experiment is on a rubber-surfaced sports field with a 21-L payload (Fig. 5). After a 2 s zero-command interval, the robot tracks a forward command of 0.8 m/s. Fig. 4 and Tab. VI show that the mean forward velocity decreases from 0.725 m/s for unloaded Base to 0.649 m/s for towing Base, whereas DRICS reaches 0.803 m/s. It reduces forward MAE from 0.158 to 0.085 m/s, a 46.2%

TABLE VI: Hardware tracking MAE over 2–10 s. Values are trial mean $\pm$ sample SD, and n is the number of trials. The lower towing forward MAE in each scene is bold.

Controller/load	n	v_x MAE $\downarrow$	v_y MAE $\downarrow$	ω_z MAE $\downarrow$
<i>Indoor: $v_x^d = 0.5$ m/s</i>				
Base, unloaded	1	0.099	0.037	0.086
Base, towing	2	0.166 ± 0.007	0.049 ± 0.003	0.108 ± 0.006
DRICS, towing	2	0.053 ± 0.002	0.050 ± 0.002	0.096 ± 0.010
<i>Outdoor: $v_x^d = 0.8$ m/s</i>				
Base, unloaded	2	0.123 ± 0.007	0.060 ± 0.003	0.129 ± 0.004
Base, towing	2	0.158 ± 0.003	0.088 ± 0.006	0.130 ± 0.004
DRICS, towing	2	0.085 ± 0.002	0.082 ± 0.012	0.115 ± 0.015

reduction, without degrading lateral or yaw tracking. Fig. 6 shows the mean forward reference to the frozen base policy is 0.962 m/s, compensating for the sustained towing resistance.

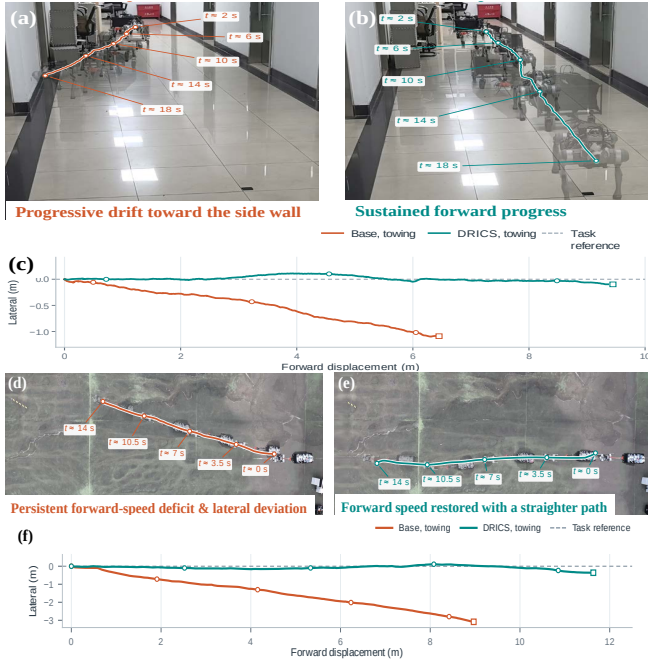


Fig. 5: **Indoor (a,b,c) and outdoor (d,e,f) loaded-cart towing.** The Base policy progressively drifts, whereas DRICS maintains sustained forward progress and a substantially straighter planar path.

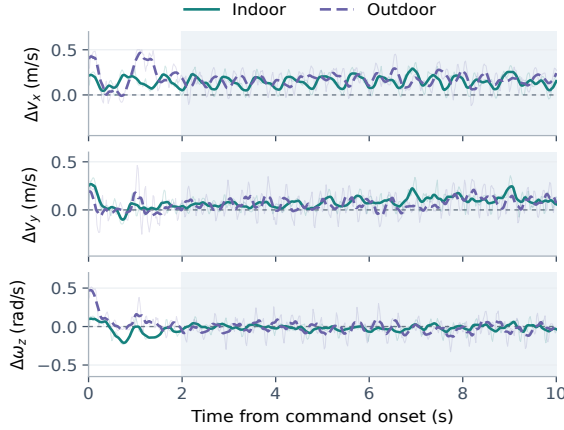


Fig. 6: **Learned command corrections in towing.** Three-axis corrections for the same DRICS trials as Fig. 4. Faint and dark curves show raw outputs and centered 0.2 s moving averages. Shading marks the 2–10 s analysis interval.

VII. CONCLUSION

We presented DRICS, a command-space disturbance-rejection framework that learns bounded 3-D command corrections through a frozen locomotion policy while retaining the original user command as the task objective. An SRB model defines the planar disturbance representation, and an ESO provides force and yaw-moment disturbance estimates. In simulation, DRICS-ESO reduces disturbed linear and yaw tracking errors by 69.1% and 62.9%, while preserving nominal performance across PPO and HIM bases. Although Unified is modestly more accurate, it requires full end-to-end retraining. In indoor and outdoor loaded-cart towing, DRICS reduces forward-velocity tracking MAE by 68.0%

and 46.2%. These results support modular reuse of pretrained locomotion skills under sustained interaction disturbances. Future work will incorporate estimator-in-the-loop training and extend DRICS to broader interactions and full-body command spaces for coupled locomotion and manipulation.

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